

CAER (Culturally-Aware Emotion Retrieval): Unified multimodal affective RAG system
Cultural context-aware emotion understanding

The existing RAG systems available retrieve information but ignore affective context, and existing emotion models ignore cultural specificity. The proposed research work fulfills both the objectives.

The Kota Factory dataset in the proposed research is the cultural testbed, the multimodal RAG (Milvus + IBM dockling) is the retrieval backbone, and your audio-to-emotion module provides the affective signal.

The paper proposes a new task called "culturally-aware affective retrieval" and benchmark it against standard RAG + standard emotion models separately. emotion is not a universal signal -it is culturally situated. Therefore, a retrieval system that indexes affective states without cultural grounding will systematically misrank documents for users whose emotional expression is culturally distinct from the training distribution.

The three model components (M1-Kanan, M2-Divyanshi, M3-Om)

M1 is your audio-to-emotion work reframed as an "affective encoder" that maps audio features (valence, arousal via wav2vec2 or similar) into a latent vector space.

M2 is your Kota Factory dataset reframed as a "cultural index" a corpus of scene-level context embeddings that capture Indian social and emotional norms.

M3 is a new scoring function you define: the Affective Retrieval Score (ARS), which combines standard cosine semantic similarity with an affective alignment weight and a cultural proximity weight.

The three falsifiable hypotheses (H1, H2, H3)

H1 says ARS will outperform pure cosine similarity on MRR@10. H2 says the performance gap will be significantly larger on culturally-specific Indian-context queries than on generic queries. H3 says audio+text fusion will outperform text-only retrieval on emotion-F1. These are the three things you run experiments to prove. If any of them fail, that is also a publishable result.

ARS scoring function definition (a weighted linear combination, a baseline comparison against standard RAG (without cultural or affective components), and an ablation study showing each component's individual contribution.

Proposed Solution:

maps each of the three theories directly to a model component: Schwarz's Affect-as-Information theory justifies M1 (the affective encoder), Mesquita's Cultural Appraisal theory justifies M2 (the cultural index from your Kota Factory dataset), and Lave's This section presents the Culturally-Aware Affective Retrieval (CAER) framework. We formalize the theoretical grounding, define the model's three functional components, state the three falsifiable hypotheses the model tests, and specify the evaluation protocol. The CAER model makes a single overarching scientific claim:

Let the following notation hold throughout this section. All vectors are real-valued. The retrieval corpus D consists of multimodal documents drawn from the KF-MED dataset

Table II. Notation Reference

Notation	Definition
q	A natural language query submitted by the user
$d_i \in D$	The i -th document in the retrieval corpus D (multimodal: text, audio, video frame)
v_q	Semantic embedding of query q (produced by a frozen LLM encoder, e.g. BGE-M3)
a_q	Affective vector of query q : 2-dimensional (valence v , arousal a), extracted from audio input
c_d	Cultural context vector of document d : extracted from KF-MED scene metadata (scene type, speaker role, social pressure index)
e_d	Emotion label embedding of document d : encoded from human-annotated emotion tags in KF-MED
$ARS(q, d)$	Affective Retrieval Score — the proposed scoring function (defined in Section III-D)
α, β, γ	Learnable scalar weights for the three ARS components ($\alpha + \beta + \gamma = 1$)
KF-MED	Kota Factory Multimodal Emotion Dataset — the curated corpus introduced in this work

Model Architecture: Three Components

The CAER framework is composed of three functional components, each independently ablatable, which together produce the ARS retrieval score. Figure 1 illustrates the end-to-end data flow.

M1 — Affective Encoder

The Affective Encoder maps raw audio input into a 2-dimensional affective vector $\mathbf{a}_q = (v, a)$, where v denotes valence (positive vs. negative affect) and a denotes arousal (calm vs. excited). We adopt the continuous dimensional model of emotion (Russell, 1980) rather than discrete categorical labels, as the former is more compatible with retrieval scoring via inner product.

Audio features are extracted using wav2vec 2.0 (Baevski et al., 2020) fine-tuned on RAVDESS and CMU-MOSI for valence/arousal regression. The KF-MED corpus provides additional fine-tuning data with Hindi-English code-switched speech, addressing the domain shift problem that arises when applying Western-trained speech emotion models to Indian media content.

$$\mathbf{a}_q = \mathbf{f}_{\text{audio}}(\mathbf{x}_{\text{audio}}) \rightarrow (v, a) \in [-1, 1]^2$$

Eq. (1): Affective encoding of audio input into valence-arousal space.

M2 — Cultural Index

In this research paper we need this to be tested with results attached

Past research theories

Schwarz's Affect-as-Information theory (1983), Mesquita's Cultural Appraisal theory (2001), and Lave's Situated Cognition framework (1988).

Cultural index calculation study

The Cultural Index is a pre-computed vector store of scene-level embeddings drawn from the KF-MED corpus. Each document d_i in the retrieval corpus is associated with a cultural context

vector $c_d \in R^k$, where k is the dimensionality of the cultural embedding space (empirically set to 128 in our experiments).

The cultural context vector encodes four metadata dimensions extracted from the KF-MED annotation schema:

- Scene type: classroom, hostel room, examination hall, counseling session, public space
- Speaker role: student, teacher, parent, peer, self-monologue
- Social pressure index (SPI): a continuous scalar $[0,1]$ derived from annotator consensus on perceived social pressure in the scene
- Affect expression norm (AEN): a binary flag indicating whether the scene context suppresses or permits overt emotional expression per Indian social norms

Cultural context vectors are produced by a fine-tuned sentence encoder (specifically, IndicBERT applied to scene description transcripts). The Cultural Index is stored in the Milvus vector database and queried in parallel with the semantic retrieval pipeline, enabling efficient nearest-neighbor lookup over the cultural space without adding latency to the generation path.

$$c_d = g_culture(scene_meta_d) \in R^{128}$$

Eq. (2): Cultural context embedding of document d from KF-MED scene metadata.

M3 — Affective Retrieval Score (ARS)

The Affective Retrieval Score is the core theoretical contribution of the CAER model. It extends the standard cosine retrieval score with two additional terms: an affective alignment term and a cultural proximity term.

$$ARS(q, d) = \alpha * \cos(v_q, v_d) + \beta * \text{sim_affect}(a_q, e_d) + \gamma * \text{sim_culture}(a_q, c_d)$$

Eq. (3): Affective Retrieval Score — the proposed scoring function.

Where:

- $\cos(v_q, v_d)$ is the standard cosine similarity between query semantic embedding and document semantic embedding. This is the term used by all baseline RAG systems and corresponds to the alpha weight.
- $\text{sim_affect}(a_q, e_d)$ is the inner product between the query affective vector a_q and the document emotion label embedding e_d , both projected into a shared affective space via a learned linear projection W_a . This measures how emotionally congruent the retrieved document is with the affective state expressed in the query.
- $\text{sim_culture}(a_q, c_d)$ is the cosine similarity between the query's cultural context vector (inferred from scene metadata if available, or imputed from the affective vector via a learned mapping) and the document's Cultural Index vector c_d . This term rewards retrieval of documents whose cultural context matches the emotional expression norm of the query context.

The three scalar weights alpha, beta, gamma are jointly optimized during training on the KF-MED corpus using a contrastive ranking objective (InfoNCE loss), with the constraint $\alpha + \beta + \gamma = 1$. Enforcing a simplex constraint ensures interpretability: the final weights indicate the relative contribution of semantic, affective, and cultural signals to retrieval precision on the target corpus.

$$L = -\log[\exp(\text{ARS}(q, d)) / \sum_j \exp(\text{ARS}(q, d_j))]$$

Eq. (4): InfoNCE contrastive ranking loss for joint optimization of alpha, beta, gamma.

Ablation Study Design

To isolate the contribution of each model component and avoid confounding the hypotheses, we define the following ablation conditions. All conditions share the same retrieval corpus (KF-MED) and the same underlying LLM backbone (LLaMA-3-8B via IBM WatsonX with Docker containerization), differing only in the scoring function.

Table III. Ablation Conditions for CAER Evaluation

System	Affect signal	Cultural index	Scoring fn.	Metrics
Standard RAG (baseline)	No	No	Cosine	MRR, F1

Text-emotion RAG	Text only	No	Cosine+emotion	MRR, F1
Audio-emotion RAG	Audio only	No	Cosine+audio	MRR, F1
CAER (proposed, full)	Audio+Text+Video	Yes (KF-MED)	ARS	MRR, F1, CP, ARS-delta

Evaluation Metrics

The following four metrics collectively evaluate whether CAER's theoretical claims hold empirically:

MRR@10 (Mean Reciprocal Rank at cutoff 10): The primary retrieval metric. For each query, $MRR@10 = 1/\text{rank_of_first_relevant_document}$, averaged over all test queries. This measures whether emotionally relevant documents are ranked highly, not just whether they appear somewhere in the result list.

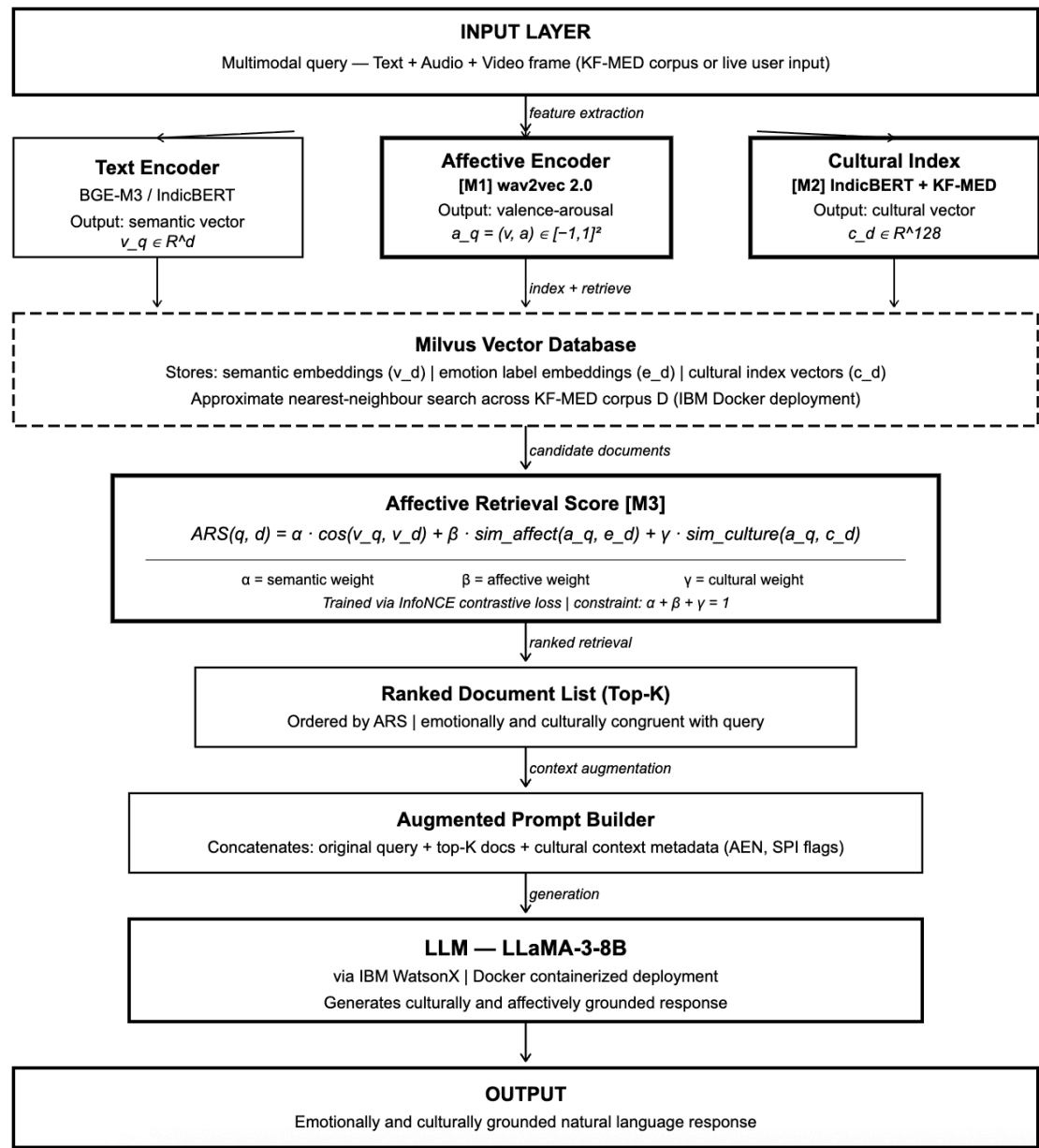
Emotion-F1: Macro-averaged F1 score over the six emotion categories (anxiety, frustration, determination, hopelessness, relief, pride) annotated in the KF-MED corpus. Computed over the top-5 retrieved documents per query to assess emotional coverage.

Cultural Precision (CP): The proportion of retrieved documents whose AEN flag (Affect Expression Norm) is consistent with the AEN of the query context. This metric directly operationalizes H2 by measuring cultural relevance independent of semantic content.

ARS-delta: The absolute improvement in MRR@10 attributable to the beta and gamma terms of the ARS function, computed by subtracting the MRR@10 of the alpha-only (cosine) condition from the full ARS condition. A positive ARS-delta on culturally-specific queries and a near-zero ARS-delta on culturally-neutral queries would confirm H2.

Situated Cognition justifies M3 (the ARS scoring function).

CAER: Culturally-Aware Affective Retrieval — Architecture Block Diagram



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Reference Annotation Guide

Ref. Range	Covers	Maps to CAER Component
[1]–[10]	RAG foundations, multimodal RAG surveys, M4-RAG	Pipeline backbone, Milvus deployment, IBM Docker context
[11]–[20]	Multimodal emotion in conversation (MERC)	KF-MED annotation schema, conversational emotion labels

[21]–[27]	Open-vocabulary MER, LLM-based emotion recognition	Open-vocabulary motivation for CAER, MER-OV benchmark gap
[28]–[35]	Speech emotion recognition, wav2vec 2.0, valence/arousal	M1 Affective Encoder — direct technical foundation
[36]–[43]	Cultural emotion, affect-as-information theory, situated cognition	M2 Cultural Index — theoretical grounding (Section III-A)
[44]–[47]	RAG + emotion / affective computing applications	Closest related work — gap analysis for novelty claim
[48]–[51]	Milvus, BGE-M3, sentence embeddings, InfoNCE	M3 ARS function — retrieval infrastructure
[52]–[56]	LLaMA, GPT-4o, Qwen-Audio, Transformer, REALM	Generation backbone and model context
[57]–[61]	IEMOCAP, MELD, RAVDESS, CMU-MOSI, MOSEI, ACE	Baseline dataset comparison for KF-MED positioning
[62]–[64]	InfoNCE loss, DPR, MER-OV evaluation	M3 training objective and evaluation metrics